

CampusShield AI

Judge Preparation & Technical Summary

A concise preparation guide based on the supplied project architecture, implementation plan, database design/implementation, technical decisions, agent tasks, and UI/UX design documents, with current official Supabase documentation used only for external verification.

1. What We Are Building

CampusShield AI is an AI-powered campus safety and smart college ERP platform. Safety is the core differentiator: students can report incidents, AI can classify and prioritize them, and security/admin teams can monitor incidents through a real-time command center and campus map.

The wider platform also covers students, faculty, attendance, exams, timetable, hostel, transport, complaints, placements, communication, parent access, visitors and audit logs.

2. Main Technology Stack

Layer	Technology	Purpose
Frontend	Next.js 14 + TypeScript	App Router, Server/Client Components, dashboards
UI	Tailwind CSS + shadcn/ui	Accessible, reusable interface components
Backend	Next.js Route Handlers	API layer and controlled mutations
Database	Supabase PostgreSQL	Relational ERP + safety data
Auth	Supabase Auth + @supabase/ssr	Cookie-based authentication
Security	RBAC + PostgreSQL RLS	Role and row-level access control
AI	Gemini 2.0 Flash	Incident classification and safety insights
Realtime	Supabase Realtime	Live incidents, SOS and emergency alerts
Map	Custom SVG	Campus-specific safety visualization

3. Most Important Demo Flow

1. Student reports an incident in natural language.
2. Gemini analyzes the report.
3. AI returns structured JSON: category, severity, confidence, summary, department, recommended actions and risk factors.
4. Backend validates the AI response.
5. Authentication and RBAC are checked.
6. The incident is written to PostgreSQL.
7. Supabase Realtime pushes the update to the command center and campus map.
8. Security/admin staff can review, assign, update and resolve the incident.

4. Critical AI Safety Point

AI assists; AI never acts directly. Gemini does not receive database credentials and does not directly modify the database. Its output goes through backend validation and authorization before a database mutation. Human operators remain in control.

5. Security Architecture

- **Authentication:** Supabase Auth with cookie-based sessions.
- **RBAC:** Roles determine which routes and actions a user can access.
- **API authorization:** Route handlers authenticate, authorize and validate before mutations.
- **RLS:** PostgreSQL Row Level Security provides database-level row restrictions.
- **Auditability:** Important mutations and incident timeline events are logged.
- **Secrets:** Service-role credentials remain server-side and are never exposed to the browser.

6. Main Features

- AI-assisted incident reporting and classification
- Safety command center
- Live campus safety map
- SOS panic/emergency workflow
- Emergency broadcasts
- Safety analytics and AI insights
- Visitor management
- Incident timeline and evidence
- Role-based dashboards
- ERP modules
- Audit logs

7. Main Judge Questions & Strong Answers

Q: What is innovative about your project?

A: We combine AI-assisted incident classification with a real-time campus safety command center. The AI converts unstructured reports into structured safety information, while humans remain responsible for consequential decisions.

Q: Why do you need AI?

A: Manual incident reports are unstructured and require people to classify, prioritize and route them. AI accelerates that process by producing category, severity, confidence, routing and recommended actions.

Q: Can AI directly change your database?

A: No. Gemini only returns structured recommendations. The backend validates the response, checks authorization and performs the database mutation.

Q: What if the AI is wrong?

A: AI output is treated as a recommendation. We validate the schema, retain confidence information and keep human operators in the loop so classifications can be reviewed or overridden.

Q: Why Gemini?

A: For this hackathon, Gemini Flash was selected for fast responses, structured JSON output, classification suitability and cost/velocity considerations.

Q: Why Supabase?

A: It gives us PostgreSQL, Auth, Storage and Realtime in one platform, reducing integration work while preserving relational modeling and database-level security.

Q: Why PostgreSQL?

A: The application has many relational entities—students, courses, attendance, incidents, departments, assignments and parents. PostgreSQL provides joins, foreign keys, constraints and transactional consistency.

Q: How does realtime work?

A: Supabase Realtime broadcasts database changes and application events over WebSockets. The command center subscribes to incident events, while security/admin views subscribe to SOS and emergency events.

Q: How is the system secured?

A: Authentication, role-based access, API authorization, PostgreSQL RLS and audit logging provide defense in depth.

Q: What if someone bypasses the frontend?

A: Security is not dependent on UI visibility. API authorization and PostgreSQL RLS enforce permissions server-side and at the database layer.

Q: Why a custom SVG campus map?

A: The environment is a fixed campus. SVG gives precise control over buildings, zones, pins and interactions without depending on a global mapping service.

Q: Why combine ERP with safety?

A: Safety information becomes more useful when connected to institutional context such as students, faculty, hostel, complaints and administration. The ERP provides that operational context.

Q: Did you train your own AI model?

A: No. We use a pretrained foundation model. Our engineering contribution is the safety schema, prompts, validation, routing logic, confidence handling and human-in-the-loop workflow.

Q: What happens if Gemini is unavailable?

A: The safety workflow should degrade safely: the incident can still be captured for human review rather than allowing an unavailable AI service to become a single point of failure.

Q: How would you evaluate AI accuracy?

A: We would build a labeled test set of historical/synthetic incidents and measure classification accuracy, severity agreement, routing accuracy and false-positive/false-negative rates, with human review for high-risk cases.

8. Ten Questions to Memorize

1. What is innovative?
2. Why AI?
3. Can AI directly act?
4. Why Gemini?
5. Why Supabase?
6. How is data secured?
7. How does realtime work?
8. What happens if AI is wrong?
9. Why a custom campus map?
10. Why ERP + safety?

9. Important Presentation Advice

Do not overclaim. Clearly distinguish between features that are implemented and demonstrated versus features that are designed/planned in the architecture. The project documents describe a comprehensive 24–48 hour hackathon plan, so credibility is stronger when the team states exactly what is live.

10. External Verification

Current official Supabase documentation confirms that RLS provides granular row-level authorization, can be combined with Supabase Auth for end-to-end security, and that service-role/secret keys must remain server-side because they bypass RLS. These points support the security explanation above.

Source basis: CampusShield AI project documents supplied in this conversation. External verification: official Supabase documentation.